Managerial Economics

Lecture 1 Fundamentals

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RENMIN BUSINESS SCHOOL

Caution

Enter at your own risk of NOT Learning Everything at this Course!

Thing One: How to Get Rich?





Thing Two: Predicting the Stock Market

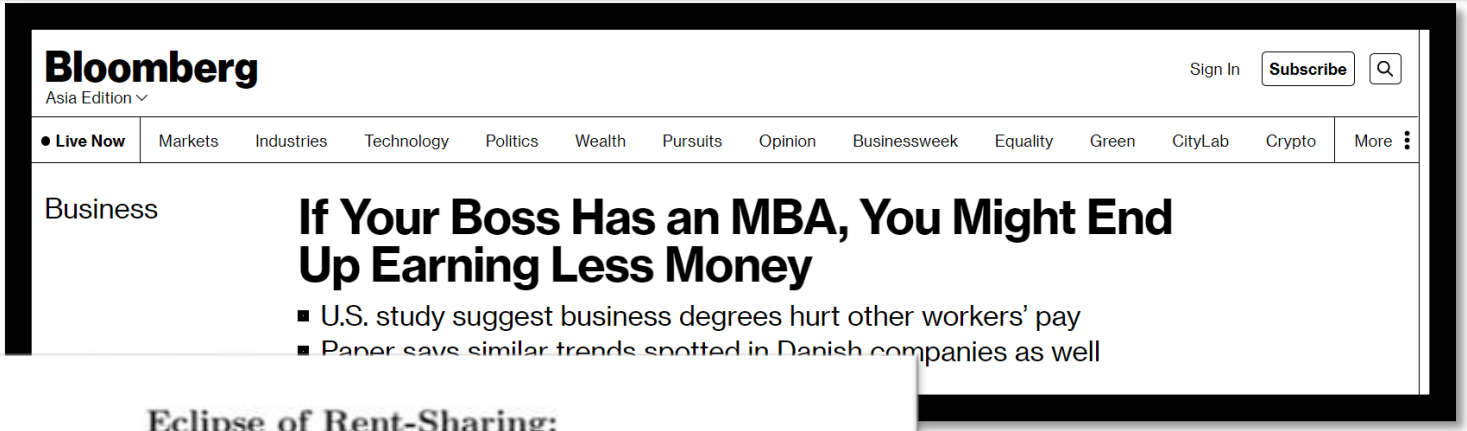




Thing Three: How to Find a Good Job?



Possible Unpleasant Consequences



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Business

If Your Boss Has an MBA, You Might End Up Earning Less Money

- U.S. study suggest business degrees hurt other workers' pay
- Paper says similar trends spotted in Danish companies as well

Eclipse of Rent-Sharing:
The Effects of Managers' Business Education on Wages and the Labor Share in the US and Denmark*

Daron Acemoglu, MIT
Alex Xi He, University of Maryland
Daniel le Maire, University of Copenhagen

March 2022

Abstract

This paper provides evidence from the US and Denmark that managers with a business degree ("business managers") reduce their employees' wages. Within five years of the appointment of a business manager, wages decline by 6% and the labor share by 5 percentage points in the US, and by 3% and 3 percentage points in Denmark. Firms appointing business managers are not on differential trends and do not enjoy higher output, investment, or employment growth thereafter. Using manager retirements and deaths and an IV strategy based on the diffusion of the practice of appointing business managers within industry, region and size quartile cells, we provide additional evidence that these are causal effects. We establish that the proximate cause of these (relative) wage effects are changes in rent-sharing practices following the appointment of business managers. Exploiting exogenous export demand shocks, we show that non-business managers share profits with their workers, whereas business managers do not. But consistent with our first set of results, these business managers show no greater ability to increase sales or profits in response to exporting opportunities. Finally, we use the influence of role models on college major choice to instrument for the decision to enroll in a business degree in Denmark and show that our estimates correspond to causal effects of practices and values acquired in business education—rather than the differential selection into business education of individuals unlikely to share rents with workers.

Takeaways: *Managers with a business degree do not share profits with their workers!*

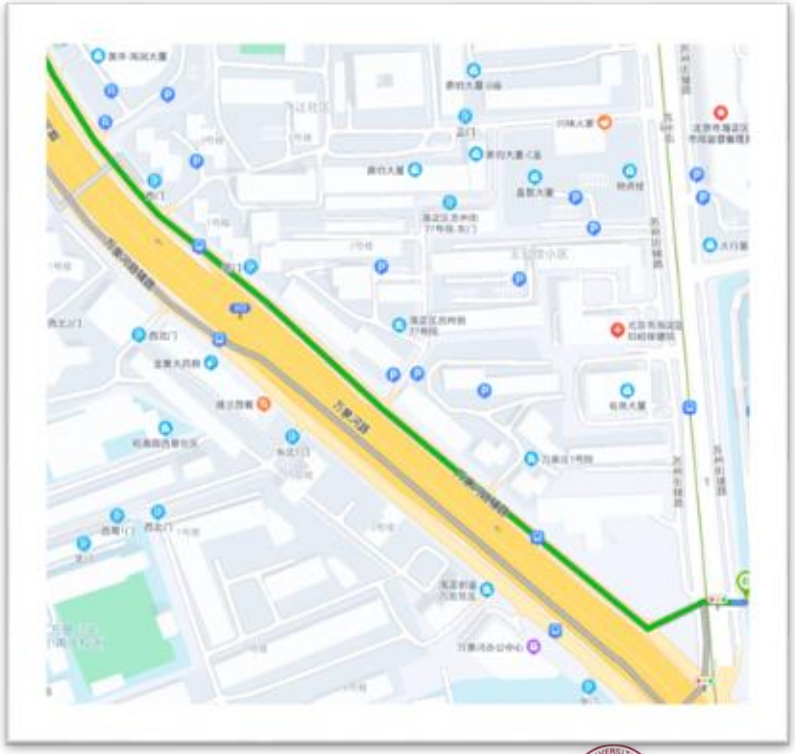
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What You will Learn?

Help Bob Choose the Right Road Map

- Bob lives in *Beijing* and plans to drive to *Shanghai* to visit his friends for the first time to celebrate the Spring Festival
- Bob uses Baidu(百度)/GaoDe(高德) map, the map provides either a *satellite view* or a *simple street view*, which view will Bob choose?





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Why We Need Economic Theory?

- "*I don't want ivory-tower theorizing; I want a practical solution!*"
  - NO cookbook formulas for real-world problems
  - Theory allows people to gain insights into complicated problems using simplifying assumptions
- Using economic theory is like using a road map
- The economic approach to understanding business *simplifies business problems* (pricing, advertising, investment…) *to their most essential components*
- It is not exaggerating that this class will provide you a way of viewing complex real-world problems and thinking of practical solutions

Not All Simplifications Are Useful

A chemist, a physicist and an economist are trapped on a desert island. All they have to eat is tinned food and no opener.

Chemist says, "I know, let's cause the salt water in the ocean to react with the tins in order to open them."

"Too slow". says the physicist. "We can focus the sun's rays through my glasses and burn our way in."

"You're going about this all wrong" says the economist. "First let's assume we have a tin opener..."



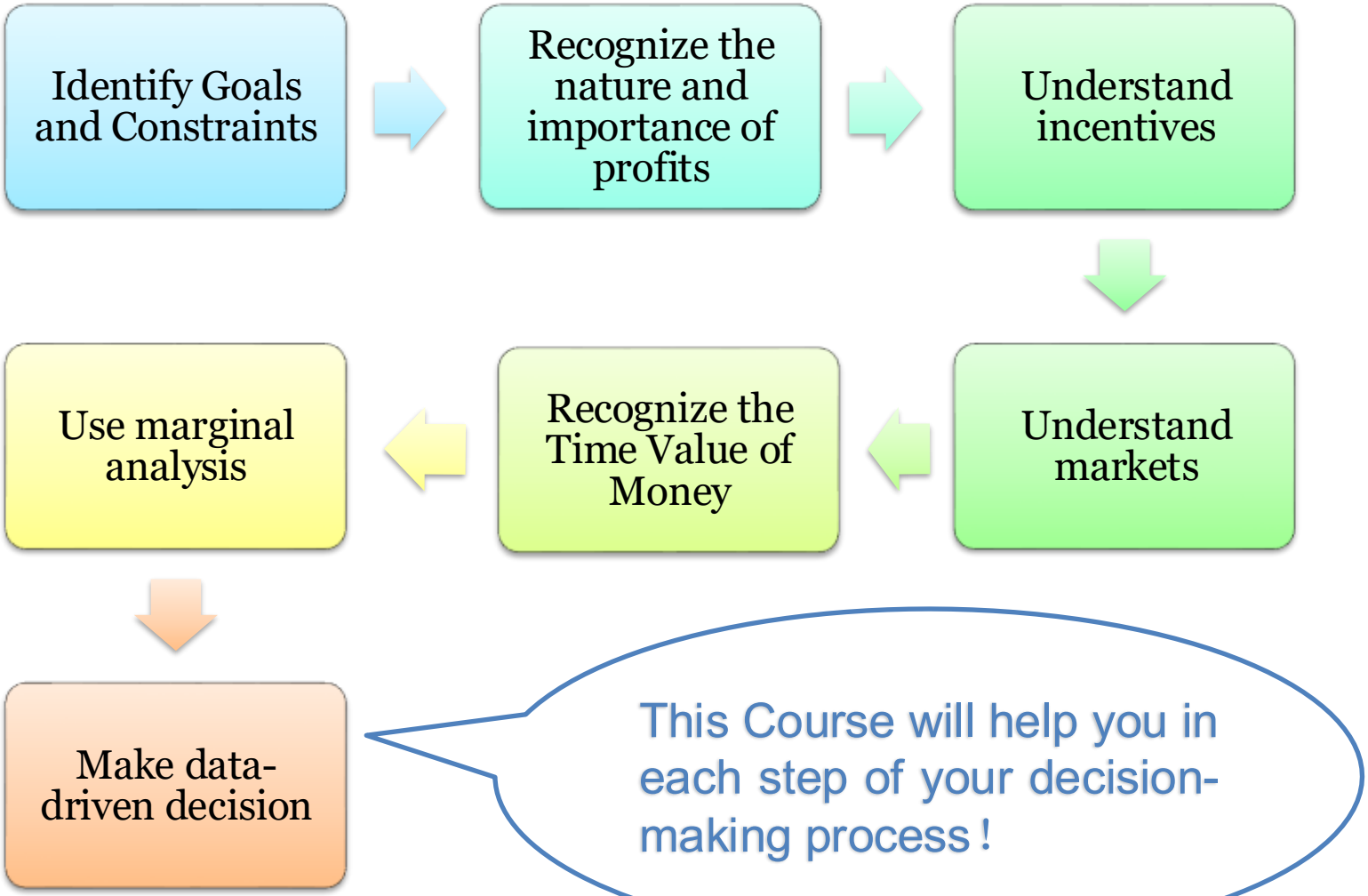
ME will utilize a large part of useful simplifications out of economics

What is Managerial Economics?

Manager + Economics = Managerial Economics

- A **manager** is a person who *directs resources /behavior of individuals* to *achieve a stated goal*.
  - Direct the efforts of others: firm, family, or a club
  - Purchase inputs to be used in production of goods and services
  - Making other decisions: product price, quality, innovation
- **Economics** is the science of *making decisions /allocating resources* in the presence of *scarce resources*
  - **Microeconomics:** individual behavior
  - **Industrial Organization:** structure of firms and industries
- **Managerial economics** is the study of how to *direct scarce resources* in the way that *most efficiently achieves a managerial goal*.
  - A firm's profits/ personal payments
  - A family's income/happiness/welfare

Principles of Effective Management



Achieving Goals with Constraints



Identify Goals and Constraints

- Almost no decisions is made free of constraints
- Identifying goals and constraints is the first step to make right managerial decision



How much time for studying ME?

- **Goal?**
- **Constraints?**

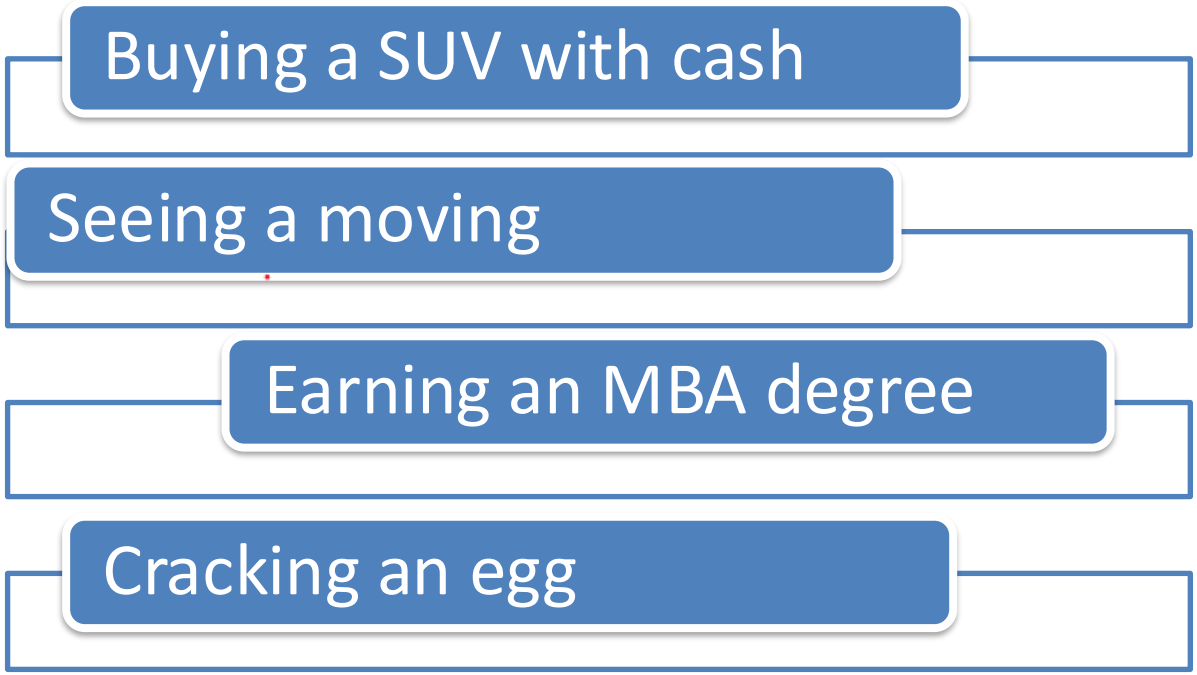


Understanding Economic Profits

Economic vs. Accounting Profits

- **Accounting Profits**
  - Total revenue (sales) minus Dollar(money) cost of producing goods or services.
  - Reported on the firm's income statement.
- **Economic Profits**
  - Total revenue minus total opportunity cost.
- The **opportunity cost (机会成本)** of using a resource includes both the *explicit (or accounting) cost* of the resource *and the implicit cost* of giving up the best alternative use of the resource
- Economic profit is no larger than accounting profit

Understanding Opportunity Costs



Are the best things in life free?

There is NO FREE lunch!

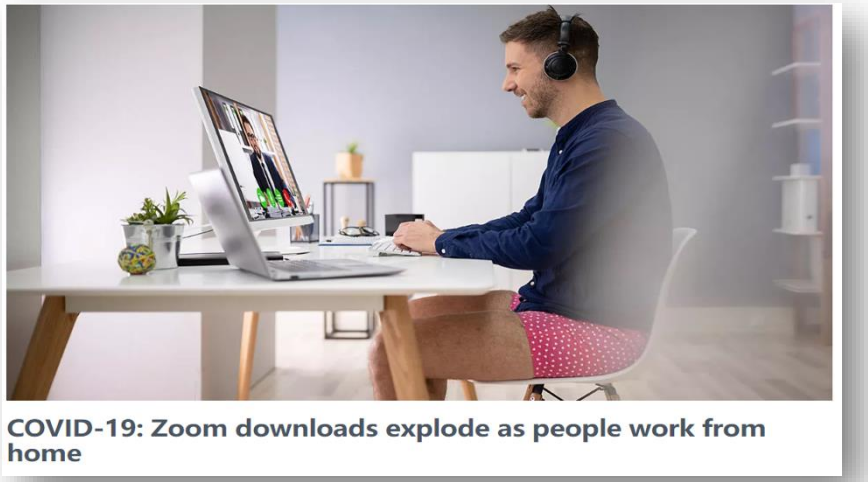
Profits as a Signal

- Adam Smith's classic line from *The Wealth of Nations*:

"It is not out of the benevolence of the butcher, the brewer, or the baker, that we expect our dinner, but from their regard to their own interest."

  - By pursuing its self-interest, a firm meets the needs of society

- Profits signal to resource holders where resources are most highly valued by society.



COVID-19: Zoom downloads explode as people work from home



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Quiz: Applying the principles

You are selling your *2016 Honda Civic*. You have already spent **2,000** Yuan on repairs.

At the last minute, the transmission dies. You can pay **1,000** Yuan to have it repaired, or sell the car "as is."

In each of the following scenarios, should you have the transmission repaired? Explain what are the opportunity costs.

A. Book value is 50,000 Yuan if transmission works, 48,500 Yuan if it doesn't

B. Book value is 50,000 Yuan if transmission works, 49,500 Yuan if it doesn't

Answers

Cost of fixing transmission = 1000 Yuan

A. Book value is 50,000 if transmission works, 48,500 if it doesn't

Benefit of fixing the transmission = (50,000–48,500) Yuan =1,500 Yuan.

It's worthwhile to have the transmission fixed.

B. Book value is 50,000 Yuan if transmission works, 49,500 Yuan if it doesn't

Benefit of fixing the transmission is only 500 Yuan.

Paying 600 Yuan to fix transmission is not worthwhile.

Answers

Observations:

- The 2,000 Yuan you previously spent on repairs is "sunk cost（沉没成本）" and is irrelevant. Don't cry for the spoilt milk.
- What matters is the cost and benefit of the *marginal* repair (the transmission).
- The change in incentives from scenario A to scenario B caused your decision to change.

Short-Run vs. Long-Run Profitability

- The **value of a firm** is the price for which the firm can be sold, which equals the present value of future profits
- Investors has uncertainty about future profits:

  $1 of profit in the future < $1 of profit now
  - **Time Value of Money (TVM)** principle

- The value of a firm is a the sum of all discounted future profits:

$$\text{Value of a firm} = \frac{\pi_1}{(1+r)} + \frac{\pi_2}{(1+r)^2} + \cdots + \frac{\pi_T}{(1+r)^T}$$

  - What determines the value of r?
  - What determines the value of T?

Get to Know the Firm Value of Tesla Corp.

Bob owns a car company *Bosla.* Bob wants to buy Tesla Corp. to strengthen its production line of electric cars.

Help Bob estimate the firm value of Tesla Corp.

Date/Period	Gross Profit ($B)	YoY Increase (%)
Q2 ending June 30, 2023		
Twelve months ending June 2023		
Annual 2022		
Annual 2021		
Annual 2020	6.63	62.94

Tesla Inc
NASDAQ: TSLA

OVERVIEW FINANCIALS

263.62 USD ▲ +3.95 (+1.52%) today
264.01 ▲ 0.39 (0.15%) After Hours · October 10, 7:59 PM EDT · Market Closed

Day Week Month Year 5 Year Max

Maximize Profits = Maximize Firm Value?

- Only in special cases, maximizing single-period profit is equal to maximizing value of the firm
  - When revenue and cost conditions in one time period are independent of revenue and costs in future time periods



Can you name some examples?

- In most scenarios, future profits correlates with today's decisions
  - A firm's employees become more productive in future periods because of learning by doing/technological progress



Impact of cutting down workers' wages on profits/firm value?

  - Current production increases production costs in the future: mining/oil extraction

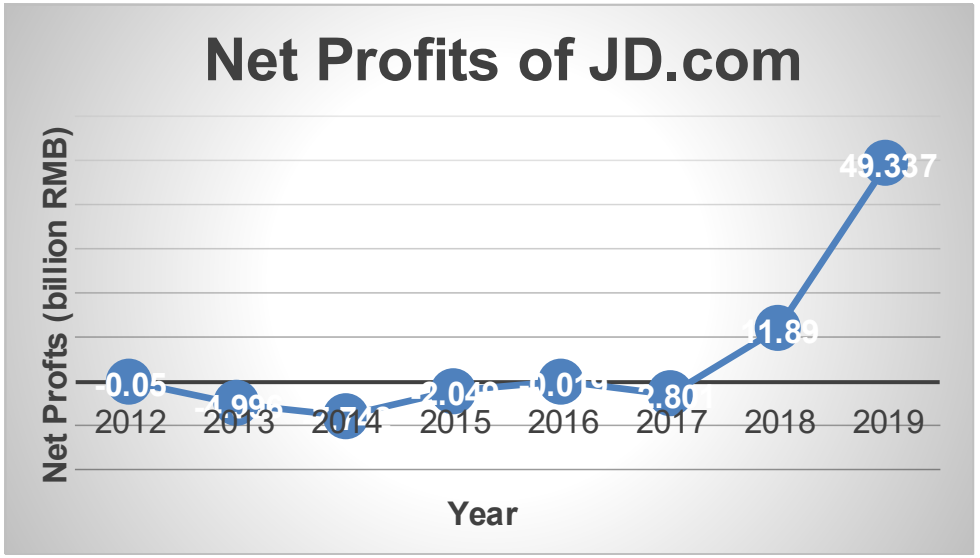
Principle If cost and revenue conditions in any period are independent of decisions made in other time periods, a manager will maximize the value of a firm (the present value of the firm) by making decisions that maximize profit in every single time period.

Long-Run Profitability

- Few close substitutes
- Strong entry barriers
- Weak rivalry within market
- Low market power of input suppliers
- Low market power of consumers
- Abundant complementary products
- Limited harmful government intervention

→ **Long-run profitability**

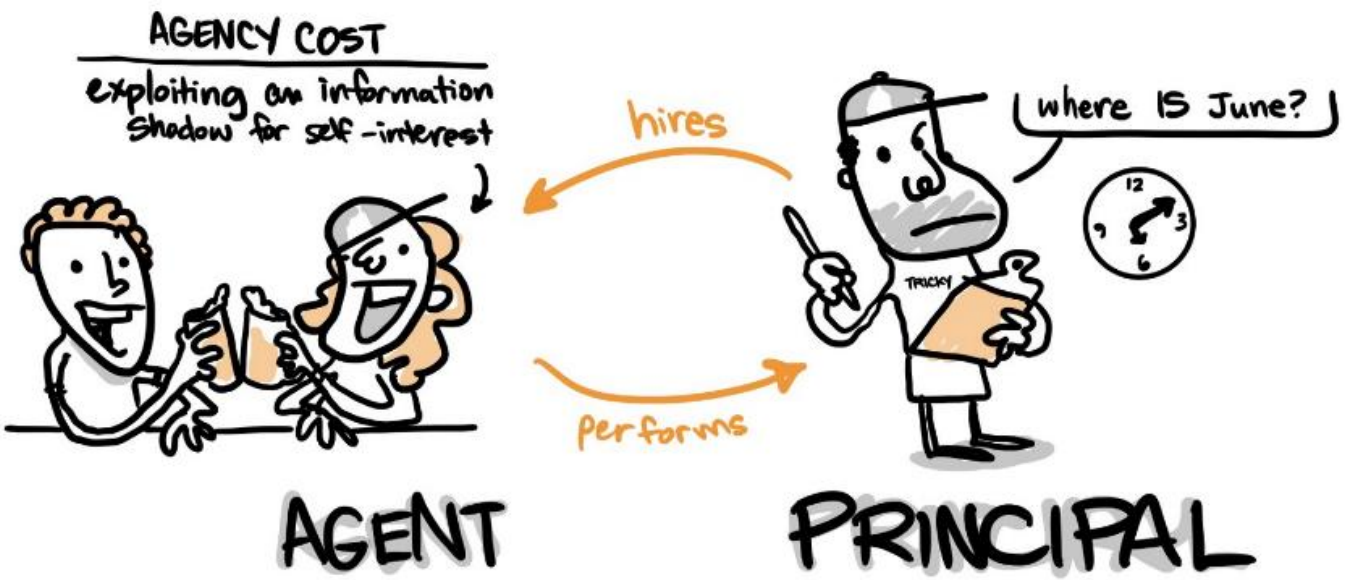




Understanding Incentives

- Do managers care more about profits or firm value?

The Principle-Agent Relationship





The Principle-Agent Problem

- Many small business owners manage their own businesses
  - Family businesses, individual business (个体户)，household
- Large business owners (*the principle*) usually delegate the control to an executive management team (*agents*)
  - **Pros:** Professional managers are more experienced and talented → increase a firm's value
  - **Cons:** Managers behave opportunistically by taking self-interested action → decrease the firm's value
- The **principle-agent problem (委托代理问题)** is a manager takes an action or makes a decision that advances the interests of the manager but reduces the value of the firm.
  - **Condition (1):** Mangers' objectives differ from the owner's
  - **Condition (2):** Monitoring the manager is too costly

Short Case Study: Help Bob Design Incentive Plans

Our old friend Bob opened a restaurant and hired a manager to run the business so he could spend time doing the things he enjoys. Recently, we asked him how his business was doing, and he reported that he had been losing money ever since the restaurant opened. When asked whether he thought the manager was doing a good job, he said, "For the 400,000 Yuan salary I pay the manager each year, the manager should be doing a good job."



What things can Bob do to incentivize his manager?

- Managers can take **hidden actions** that cannot be observed by owners for any feasible amount of monitoring effort
  - **Moral hazard** is a type of principle-agent problem; A situation when managers take hidden actions that harm the owners' interests.

Understand Markets

What is a Market?

- Market is perhaps the most important invention of the humankind
- A market is <u>any arrangement through which buyers and sellers exchange goods or services</u>
  - A wide concept beyond one's usual imagination
- Markets are arrangements to reduce transaction costs (costs of searching and purchasing)







Different Market Structures

- **Market structure** is a set of market characteristics that determines the economic environment in which a firm operates.
  - *The number and size of the firms operating in the market*
  - *The degree of product differentiation among competing producers*
  - *The likelihood of new firms entering a market when incumbent firms are earning economic profits*

Different Market Structures

Market Structure	Definition
Perfect Competition	A large number of relatively small firms selling undifferentiated products (rice, coal)
Monopoly	A single firm produces a product for which no close substitutes are available (electricity, oil extraction)
Monopolistic Competition	a large number of firms that are small relative to the total size of the market produce differentiated products without entry barriers (toothpaste, clothes).
Oligopoly	a few firms produce most or all of the market output, so any one firm's pricing policy will affect other firms (smartphones, cars).



What is the market structure in the industry you worked?

Use Marginal Analysis

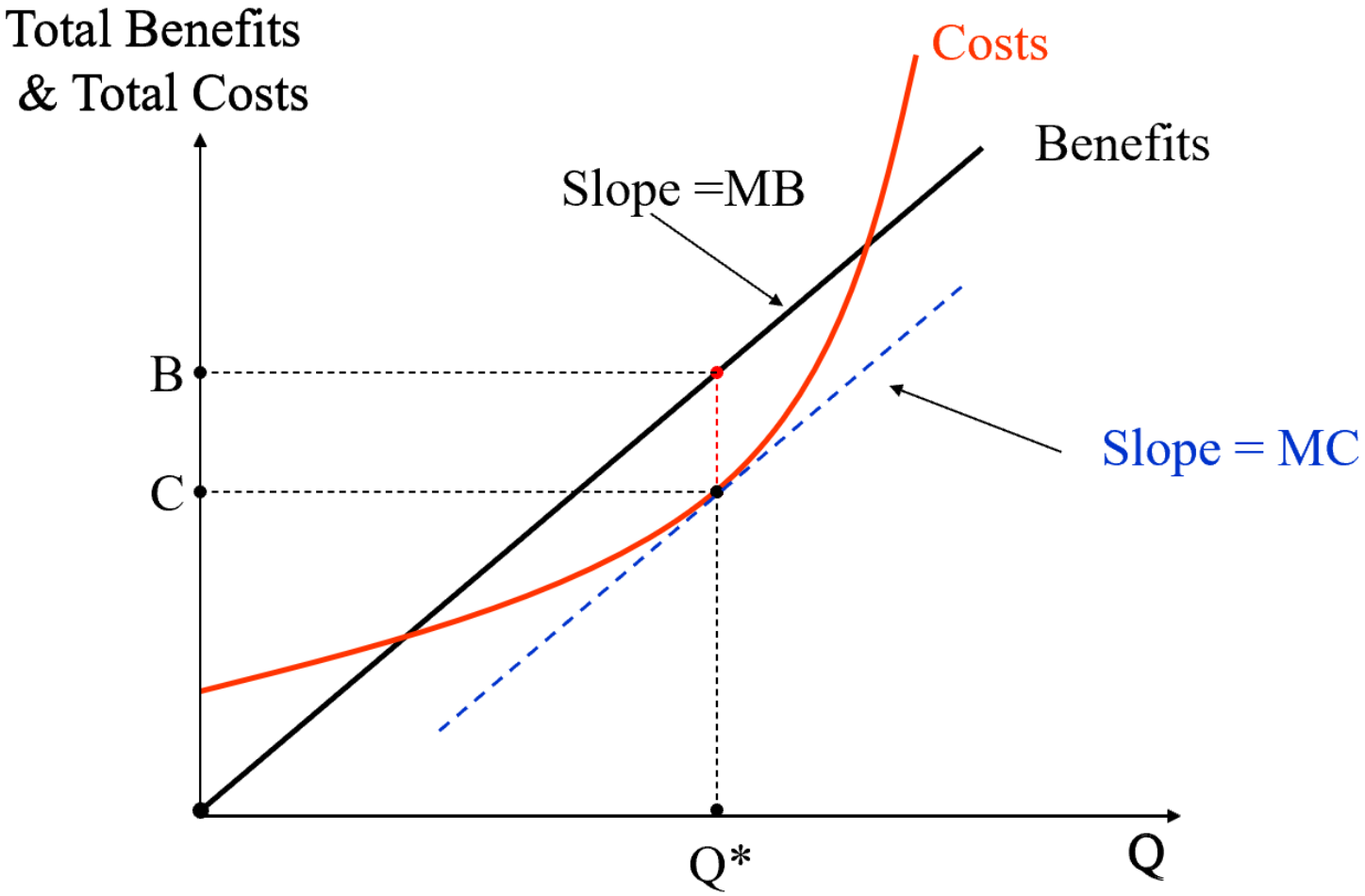
Rational People Think at the Margin

What is Marginal Analysis

- Control/Choice Variable Examples:
  - Output / Price /Product Quality/ Advertising /R&D
- **Basic Managerial Question:** How much of the control variable should be used to maximize net benefits?
- Profits = Revenue – Costs *or* Net Benefits = Total Benefits - Total Costs
  - **Marginal Benefit (MB):** *Change in total benefits from a change in the choice variable*
  - **Marginal Costs (MC):** *Change in total costs arising from a change in the choice variable*

Marginal Principle

- **MB=MC** maximizes the profits/net benefits/firm value



Real-Life Case: Marginal Analysis for Airlines Company

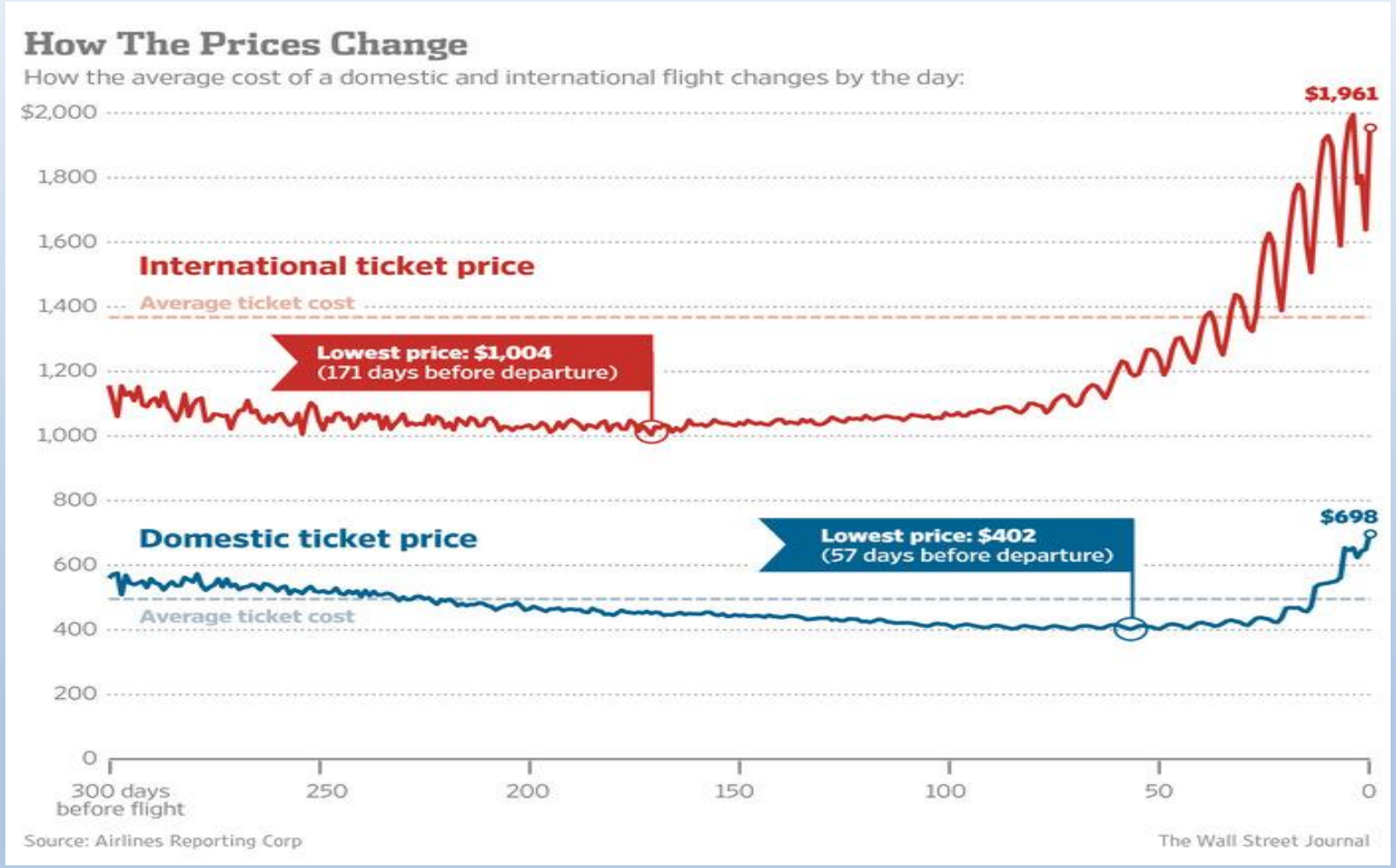
- In 1980s, Robert Crandall introduced an innovative pricing model and seat allocation system to maximize the total revenue for each seat.



Robert Crandall,
American Airlines
President and CEO, 1985-1998

- A significant portion of an airline's operating **costs are fixed**, *such as the purchase or lease of aircraft, employee salaries, aircraft maintenance, and fuel expenses.*
- **Variable costs**, like *in-flight meals and beverages, baggage handling*, etc., usually constitute a relatively small percentage of the total costs.
- As long as the ticket price exceeds the marginal cost (primarily the cost of the meal), even if this price is much lower than the regular price, the seat should be sold to increase revenue.

Real-Life Ticket Prices



Case Study: Pricing Strategy at the Hilton Hotel

Background:

The Hilton Hotel is a luxury property with **200** rooms, located in the heart of a major city. The hotel primarily caters to business travelers during the week and leisure travelers on weekends. It has a consistent occupancy rate of **85%** during weekdays but only **60%** over the weekends.

Challenge:

The hotel's management is considering a change in its pricing strategy **to boost weekend occupancy rates without compromising revenue**. They have observed *that leisure travelers are price-sensitive, unlike their business counterparts*. They've approached you, the hotel's lead economist, to assist in devising a new pricing strategy.

Summary

- *Managerial Economics* offers a way of thinking managerial problems
- The core economic theory is simple yet powerful, while the right application of theory needs training and practicing
- Core concepts and tools:
  - *Opportunity costs and economic profits*
  - *Short-run vs. long-run profits*
  - *Various market structure and pricing power*
  - *Marginal analysis*